Renewable Energy 275 (2026) 126304

Contents lists available at ScienceDirect

Renewable Energy

journal homepage: www.elsevier.com/locate/renene

Synergistic enhancement of methane yield and odor control in pig manure anaerobic digestion via micro-aeration pretreatment

Hong Zhao [a], Xiao-Bo Zhou [a], Shu-Han Zhang [a], Fan Liu [a], Teng-Xia He [a,c,*], Guang-Liang Tian [b], Xi-Wen Liang [a], Man-Man Zhang [a], Yi-Xiao Liao [a]

[a] Key Laboratory of Plant Resource Conservation and Germplasm Innovation in Mountainous Region (Ministry of Education), Guizhou Key Laboratory of Agricultural Microbiology, College of Life Sciences, Guizhou University, Guiyang, Guizhou, China
[b] Institute of New Rural Development, Laboratory of Pollution Control and Resource Utilization Technology for Mountainous Livestock and Poultry Farming, Guizhou University, Guiyang, Guizhou Province, China
[c] Key Laboratory of Karst Georesources and Environment (Guizhou University), Ministry of Education, Guizhou University, Guiyang, Guizhou, China

ARTICLE INFO

Keywords:
Pig manure
Malodorous gases control
Micro-aeration pretreatment
Anaerobic digestion
Microbial community structure

ABSTRACT

The widespread adoption of anaerobic digestion for pig manure treatment is limited by malodorous gas emissions and low energy recovery efficiency. This study systematically examined the effects of micro-aeration pretreatment over a range of intensities (0-13.9 mL/g(TS)/d) on odor abatement and digestion performance. Micro-aeration reduced emissions of eight key malodorous gases, including sulfide (H_2S), ammonia (NH_3), methanethiol (CH_4S), and ethanethiol (C_2H_6S), by 14.0-100%. Complete sulfide mineralization was achieved at intensities ≥9.3 mL/g(TS)/d. Mitigation of nitrogen- and sulfur-containing compounds was likely driven by selective enrichment of facultative aerobic consortia (e.g., unclassified_f_*Bacillaceae* and unclassified_c_*Gammaproteobacteria*) and the upregulation of sulfur oxidation and nitrogen transformation pathways. Methane production increased by 37.6-95.5%, owing to enhanced hydrolysis and acetogenesis resulting from upregulated fatty acid β-oxidation and acetyl-CoA synthesis. Moreover, syntrophic bacteria (*Syntrophomonas*) and key methanogens (*Methanothrix* and *Methanosarcina*) were selectively enriched. An aeration intensity of 4.6 mL/g(TS)/d was identified as optimal, as it balanced effective odor control and maximal methanogenic activity. These findings provide a theoretical foundation and a viable strategy for improving energy recovery and environmental sustainability in swine manure management.

1. Introduction

The rapid expansion of the global pig industry has led to a severe accumulation of manure, presenting a severe environmental management challenge [1]. Consequently, developing efficient treatment and resource recovery strategies for pig manure is essential for environmental protection and sustainable agricultural development [2]. Although rich in organic matter and nutrients, pig manure is characterized by an imbalanced carbon-to-nitrogen ratio [3]. Its improper disposal poses significant environmental risks, such as water eutrophication, soil contamination with heavy metals, and atmospheric pollution [4]. Among these concerns, the emission of intense and complex malodorous gases during microbial degradation represents one of the most immediate and perceptible hazards [5]. The emission of these malodorous gases not only creates a public nuisance and restricts the siting of farming and processing facilities, but also poses health risks to onsite workers [6].

Anaerobic digestion (AD) has been widely recognized as a promising technology for treating pig manure, achieving simultaneous stabilization, volume reduction, and energy recovery in the form of methane (CH_4) [7]. Nevertheless, the traditional anaerobic digestion process is often plagued by inherent limitations, such as low CH_4 yield and the persistent release of substantial amounts of malodorous gases [8,9]. Sulfur and nitrogen-containing compounds are key contributors to these foul-smelling gases [10,11]. These compounds are formed due to the incomplete microbial degradation of sulfur-containing amino acids and nitrogen-rich organic matter in pig manure under anaerobic conditions [12]. These malodorous gases not only impeded the broader adoption of AD technology but also compromises the safe handling and utilization of its by-products [13].

* Corresponding author. Zhijuan East Road, Huaxi, Guiyang, Guizhou 550025, China.
E-mail address: txhe@gzu.edu.cn (T.-X. He).

https://doi.org/10.1016/j.renene.2026.126304
Received 14 January 2026; Received in revised form 12 August 2026; Accepted 13 August 2026
Available online 20 August 2026
0960-1481/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

The micro-aeration pretreatment has gained significant attention as an economically viable method for controlling malodorous gases at the source [14]. The core principle involves creating a micro-aerobic environment through controlled oxygen introduction, thereby directly suppressing obligate anaerobic odor-producing bacteria, such as sulfate-reducing bacteria [15]. In such an environment, facultative and aerobic bacteria are promoted to oxidize odorous precursors into odorless or low-odor substances [16]. Concurrently, this approach can regulate nitrogen conversion and mitigate ammonia volatilization [17]. For example, microorganisms can facilitate the simultaneous transformation and removal of nitrogenous and sulfurous malodorous gases by converting ammonia (NH_3) to nitrate and oxidizing volatile sulfur compounds to sulfate [18,19]. However, during the process of micro-aeration pretreatment, a large number of aerobic or facultative aerobic microorganisms are enriched in the aerobic environment, which maybe not suitable for anaerobic fermentation system [20]. While oxygen can suppress the formation of malodorous compounds, it also oxidizes labile organic carbon to CO_2, resulting in irreversible carbon loss and a potential reduction in methane yield [15]. Furthermore, current odor evaluation systems remain incomplete. Most studies predominantly focus on H_2S and NH_3, while neglecting other significant malodorous compounds such as trimethylamine (C_3H_9N), methanethiol (CH_4S), ethanethiol (C_2H_6S), dimethyl disulfide ($C_2H_6S_2$), and carbon disulfide (CS_2). Additionally, few studies have systematically quantified, within a unified framework, the relationship between aeration intensity, organic matter loss, and terminal methane production. Therefore, the effects of micro-aeration treatment on malodorous gases control, organic matter loss, and methane production urgently require examination.

Most existing studies only analyze H_2S and NH_3 emissions and overlook critical odorants including trimethylamine, methanethiol, ethanethiol, dimethyl disulfide and carbon disulfide. Furthermore, no unified experimental system has yet quantified the coupled interplay of aeration intensity, organic matter mineralization and final methane output. Collectively, the scarcity of research in these areas demonstrates an urgent need to clarify how micro-aeration simultaneously regulates odor release, carbon loss, and methanogenic performance.

Given the aforementioned challenges, this study aims to (1) systematically quantify the release of key malodorous gases (NH_3, C_3H_9N, H_2S, CH_4S, C_2H_6S, $C_2H_6S_2$, CS_2) and overall odor levels under seven micro-aeration intensities (0-13.9 mL/g(TS)/d; (2) evaluate the long-term anaerobic digestion performance of pretreated manure and establish a quantitative relationship among aeration intensity, carbon loss, and CH_4 yield; and (3) elucidate how aeration intensity drives microbial community dynamics, including diversity, assembly processes, keystone taxa, and functional potential. These findings will provide an integrated scientific basis for the precise optimization of micro-aeration pretreatment in pig manure AD.

2. Materials and methods

2.1. Raw materials and pretreatment

Fresh pig manure was obtained from a large-scale farm in Xiuwen County, Guiyang, Guizhou Province, and promptly stored at 4 ± 1 °C. The basic physicochemical properties of the raw manure are summarized in Table S1.

2.2. Micro-aeration pretreatment system

The micro-aeration pretreatment system is provided in Fig. S1. Pig manure with a total solids (TS) content of 22.5% was subjected to static micro-aeration for 20 days. Seven aeration intensity gradients were applied: 0 (control), 2.3, 3.5, 4.6, 5.8, 9.3, and 13.9 mL/g(TS)/d, each in triplicate. During pretreatment, samples were collected every 5 days (0, 5, 10, 15, and 20 days) for analysis. Dissolved oxygen (DO), oxidation-reduction potential (ORP), pH, and electrical conductivity (EC) of the fermentation slurry were monitored in situ using a HQ40d portable analyzer (HACH, USA), while gas samples were collected for subsequent analysis.

2.3. Sequential batch AD system

Sequential batch AD was presented in Fig. S2. Pig manure pretreated for 20 days at three aeration intensities (0, 4.6, and 13.9 mL/g(TS)/d) was used as the substrate for AD assessment. For each treatment, the total solids concentration was adjusted to 10%, maintaining a working volume of 150 mL per bottle. Digestion was conducted at 35 ± 1 °C under anaerobic conditions for 48 days. Before incubation, each bottle was purged with high-purity nitrogen for 2 min to remove residual oxygen and ensure obligate anaerobiosis. Biogas was collected in Tedlar gas bags (DuPont, USA), which were equipped with two valves (one for inlet, one for sampling). At 24 h intervals, the gas volume was measured with a gas-tight syringe via the sampling valve. After measurement, the bag was evacuated and reconnected to the bottle. Gas volumes were normalized to standard temperature and pressure (0 °C, 1 atm). All experiments were carried out in triplicate.

2.4. Analytical methods

CH_4 was quantified using a GC-1120 gas chromatograph (GC, Shanghai, China) equipped with a thermal conductivity detector (TCD) and a Porapak Q packed column (2 mm × 3 mm i.d.). High-purity N_2 (99.999%) served as the carrier gas at a flow rate of 30 mL/min. The injector, column oven, and detector temperatures were set to 100 °C, 80 °C, and 150 °C, respectively. Sulfur-containing malodorous gases, including H_2S, CH_4S, C_2H_6S, $C_2H_6S_2$, and CS_2, were measured on the same GC system equipped with a flame photometric detector (FPD). Separation was achieved on a DB-Sulfur capillary column (30 m × 0.25 mm × 0.25 μm), with N_2 carrier gas at 1.0 mL/min. The FPD was operated with hydrogen (75 mL/min) and air (100 mL/min) as combustion gases. The column temperature program was as follows: initial 40 °C held for 3 min, ramped at 8 °C/min to 180 °C, and held for 5 min. The injector and detector temperatures were 180 °C and 220 °C, respectively. NH_3 and C_3H_9N were determined using the same GC fitted with a nitrogen-phosphorus detector (NPD) and a DB-5 capillary column (30 m × 0.25 mm × 0.25 μm). To minimize NH_3 adsorption, a deactivated liner was installed, and the column was preconditioned with NH_3 standards prior to analysis. The N_2 carrier gas flow was maintained at 1.2 mL/min. The temperature program was set as follows: 60 °C for 2 min, increased at 10 °C/min to 160 °C, and held for 4 min. The injector and detector temperatures were 180 °C and 200 °C, respectively. All gas measurements were calibrated using certified standard gas mixtures, with all calibration curves yielding correlation coefficients (R^2) greater than 0.999. Each sample was analyzed in triplicate, and the relative standard deviations (RSD) were consistently below 5%.

TS and volatile solids (VS) were determined gravimetrically. Specifically: approximately 10 g of homogenized sample was dried in a pre-weighed crucible at 105 °C to constant weight for TS determination. The dried residue was then ashed in a muffle furnace at 550 °C for 2 h, cooled in a desiccator, and reweighed to calculate VS content. Odor concentration was determined by dynamic olfactometry and expressed as ouE/m^3, which is equivalent to the dilution-to-threshold (D/T) ratio. This value was calculated as the dilution factor at which the gas sample was no longer detectable by50% of the panelists: ouE/m^3 = V_{Total}/V_{Sample}. TS and volatile solids (VS) were determined gravimetrically. Approximately 10 g of homogenized sample was dried in a pre-weighed crucible at 105 °C to constant weight for TS determination. The dried residue was then ashed in a muffle furnace at 550 °C for 2 h, cooled in a desiccator, and reweighed to calculate VS content.

2.5. Microbial community analysis

Fresh pig manure without pretreatment (CK group) and manure pretreated at aeration intensities of 0 (A0 group), 4.6 (A4.6 group), and 13.9 mL/g(TS)/d (A13.9 group) for 20 days were subjected to 16S rRNA high-throughput sequencing (Meiji Biomedical Technology Co., Ltd., China). Total genomic DNA was extracted using the FastDNA Spin Kit for Soil (MP Biomedicals, USA) and assessed for quality via 1% agarose gel electrophoresis. The V3-V4 region of the bacterial 16S rRNA gene was amplified in triplicate with primers 341F/806R and TransStart FastPfu DNA polymerase on an ABI GeneAmp 9700 PCR System (Applied Biosystems, USA). Amplified products were pooled, purified, and quantified (CFX Connect&T100, Bio-Rad System, USA) before library construction using the TruSeq DNA PCR-Free Kit (Illumina, USA). Sequencing was performed on an Illumina NovaSeq 6000 platform.

Raw reads were processed through Fastp (v0.23.4) and FLASH (v1.2.11) for quality control and assembly. Amplicon sequence variants (ASVs) were generated using DADA2 in QIIME2 (v2024.5) and taxonomically annotated against the SILVA database (v138). Microbial diversity (α- and β-diversity), community structure, and biomarker identification (LEfSe) were analyzed with QIIME2 and R packages (vegan, ggplot2). Functional prediction of metabolic pathways was performed with PICRUSt2 based on KEGG orthology. All bioinformatics analyses were conducted on the Majorbio Cloud Platform (www.majorbio.com).

2.6. Statistical analysis

All experimental data were processed initially in Microsoft Excel 2019. Analysis of variance (ANOVA) was performed using SPSS software, with post-hoc multiple comparisons conducted via Duncan's test at a significance level of $p < 0.05$. Key Figures were prepared using Origin 2024 software.

3. Results and discussion

3.1. Malodorous gases production

3.1.1. Nitrogen containing gases

The effects of different aeration intensities on malodorous gases production during AD are shown in Figs. 1 and 2. Fig. 1A illustrated the influence of aeration intensity on NH_3 release. Except for the 3.5 mL/g (TS)/d treatment, which exhibited a slightly higher NH_3 concentration than the unaerated control (0 mL/g(TS)/d) on day 5. All aerated treatments produced significantly lower NH_3 levels throughout the 20-day incubation period. This observation is consistent with previous studies demonstrating that low oxygen concentrations can markedly inhibit NH_3 generation [21]. In all aeration groups, NH_3 was undetectable until day 20. Higher aeration intensities (9.3 and 13.9 mL/g(TS)/d) notably shortened the NH_3 release period by 5 and 10 days, respectively. Compared with the control, NH_3 production on day 10 was reduced by 63.2%, 65.8%, 58.7%, 64.9%, 83.4%, and 96.8% in the 2.3, 3.5, 4.6, 5.8, 9.3, and 13.9 mL/g(TS)/d groups, respectively. This significant reduction in NH_3 emissions is primarily attributed to the oxygen introduced during aeration, which promotes the activity of aerobic nitrifying bacteria [22]. These microorganisms oxidize ammonium to nitrate via nitrification and partially assimilate NH_4^+ into biomass, thereby fundamentally mitigating the accumulation and volatilization of NH_3 [23]. However, digestate contains relatively high levels of inorganic nitrogen [17]. Direct application of this digestate to agricultural land may lead to soluble nitrogen being transported via surface runoff into water bodies, triggering eutrophication [24]. Therefore, it is recommended to (1) apply digestate at quantity matched to crop nitrogen demand, (2) recover nitrogen from digestate through pretreatment, (3) combine digestate with straw to balance the C/N ratio and enhance microbial nitrogen transformation, and (4) establish field buffer strips to absorb excess nitrogen [25]. These measures can effectively prevent excessive soluble nitrogen in digestate from entering water bodies and causing eutrophication.

The production pattern of C_3H_9N is shown in Fig. 1B. From day 5 to day 20, C_3H_9N yields in treatment groups with aeration intensities ≥4.6 mL/g(TS)/d were consistently lower than those in the control group. On day 10, C_3H_9N production in the 4.6, 5.8, 9.3, and 13.9 mL/g

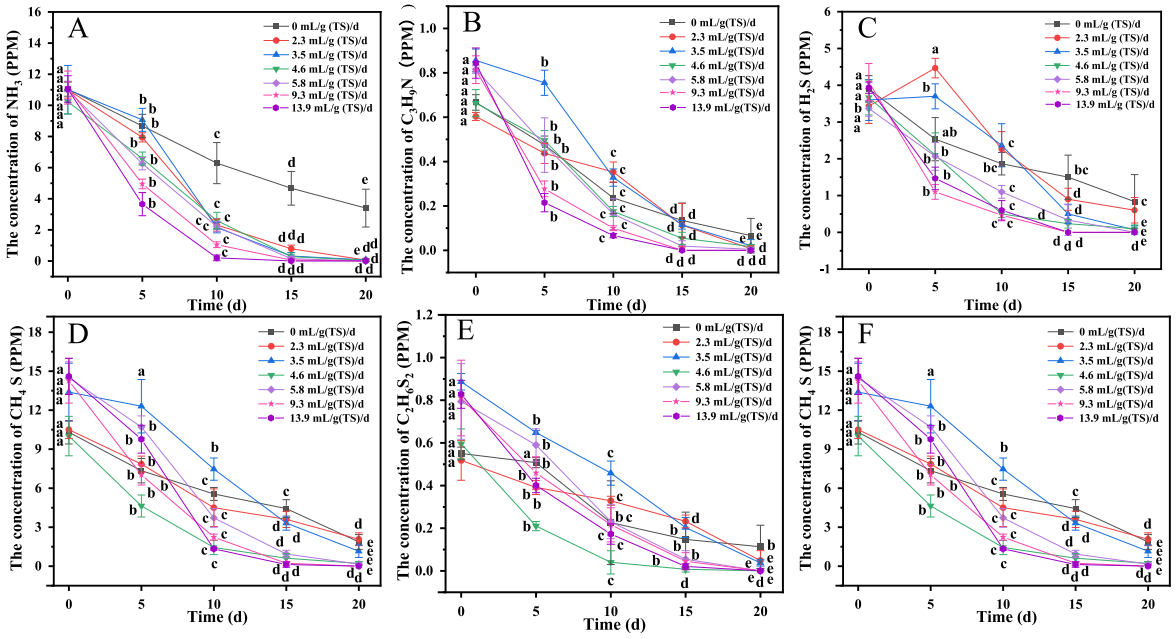


Fig. 1. Effect of micro-aeration on the production of nitrogen and sulfur related malodorous gases. A: Ammonia (NH_3), B: hydrogen sulfide (H_2S), C: trimethylamine (C_3H_9N), D: methanethiol (CH_4S), E: ethanethiol (C_2H_6S) and F: dimethyl disulfide ($C_2H_6S_2$).

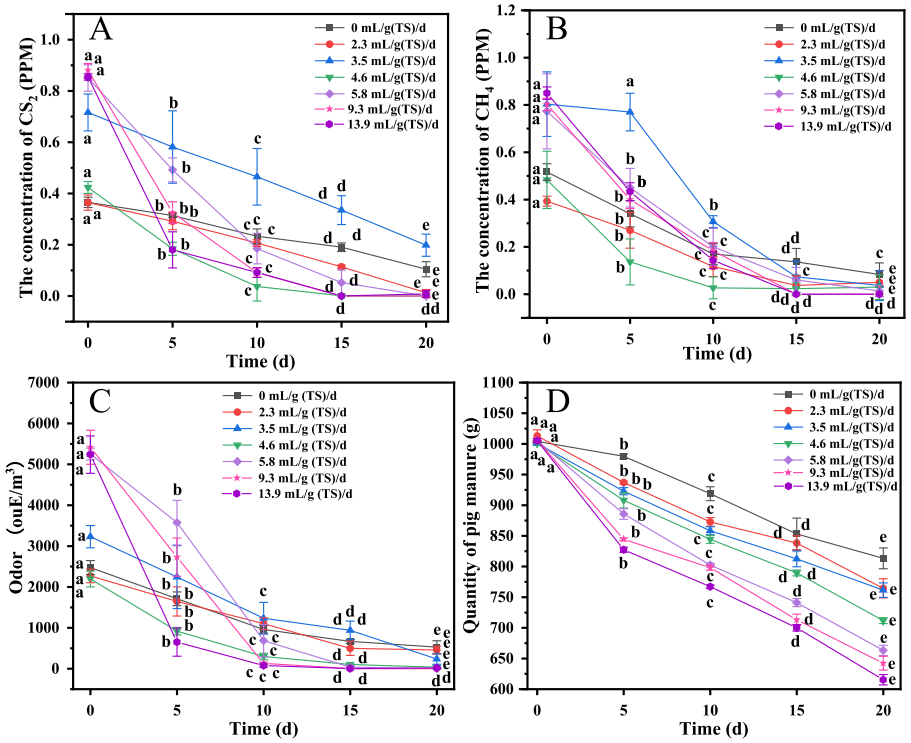


Fig. 2. Effect of micro-aeration on carbon disulfide (A: CS_2), odor (B), methane (C: CH_4) production (gas emissions from the micro-aeration pretreatment unit) and quantity of pig manure (D).

(TS)/d treatment groups decreased by 25.8%, 30.8%, 57.7%, and 71.9%, respectively, compared to the control. These results suggest that C_3H_9N formation is more effectively suppressed at higher aeration intensities. Aeration to disrupt obligate anaerobic conditions, likely inhibiting the activity of obligate anaerobes, including specific hydrolytic bacteria and methanogenic archaea involved in the anaerobic degradation of nitrogenous compounds and C_3H_9N generation [26]. Simultaneously, the introduction of oxygen activates aerobic and facultative anaerobic microorganisms. These organisms compete with anaerobic bacteria for carbon and nitrogen substrates, thereby redirecting metabolic pathways and reducing the production of odorous intermediates such as trimethylamine C_3H_9N [27].

3.1.2. *Sulfur-containing gases*

With prolonged pretreatment time, H_2S production in all treatment groups exhibited a declining trend (Fig. 1C). Notably, during days 5-20, H_2S levels in groups with aeration intensities ≥4.6 mL/g(TS)/d were significantly lower than those in the non-aerated control. By day 20, all aerated groups showed reduced H_2S production relative to the control. On day 15, H_2S emissions decreased by 40.0%, 66.7%, 84.5%, 77.8%, 100%, and 100% in the 2.3, 3.5, 4.6, 5.8, 9.3, and 13.9 mL/g(TS)/d groups, respectively, compared to the control. These results confirm that aeration pretreatment effectively suppresses H_2S generation during subsequent anaerobic treatment of pig manure. H_2S is primarily produced by obligate anaerobic sulfate-reducing bacteria (SRB), which utilize sulfate as an electron acceptor under anaerobic conditions [28]. Oxygen introduced during aeration strongly inhibited the metabolic activity of SRB and enriched sulfur-oxidizing bacteria (SOB), which might be one of the key reasons for the reduction of H_2S [29]. However, during the early pretreatment phase (5 and 10 days), H_2S production in the low-intensity aeration groups (2.3 and 3.5 mL/g(TS)/d) exceeded that of the control. This may be attributed to weak aeration stimulating aerobic or facultatively anaerobic microorganisms to decompose complex sulfur-containing organics into simpler reduced sulfur compounds, which are more readily utilized by SRB [30]. Although SRB activity is partially inhibited, increased substrate availability could temporarily promote H_2S production [31]. No H_2S was detected in the high-intensity aeration groups (9.3 and 13.9 mL/g(TS)/d) after day 15, indicating that higher aeration not only reduced H_2S formation but also completely halted its release.

CH_4S production was significantly inhibited in all aeration treatment groups, with the inhibition degree positively correlated with aeration intensity (Fig. 1D). On day 5, CH_4S concentrations decreased by 37.0% in the 4.6 mL/g(TS)/d group and by 47.0% in the 13.9 mL/g(TS)/d group compared to the control. By day 10, CH_4S levels in aerated groups ranged from 13.0% to 76.1% of the control, with reductions to 25.8% and 23.9% observed in the 4.6 and 13.9 mL/g(TS)/d groups, respectively. Complete elimination of CH_4S was achieved by day 20 in the 9.3 and 13.9 mL/g(TS)/d treatments. These results indicate that aeration pretreatment accelerates the conversion and removal of CH_4S, with complete inhibition achieved at an intensity threshold of ≥9.3 mL/g(TS)/d. This effect may be attributed to the enrichment of facultative and aerobic microorganisms, which oxidize CH_4S to odorless dimethyl disulfide or further mineralize it to sulfate within local micro-aerobic micro-niches formed during micro-aeration pretreatment [4].

The variation patterns of C_2H_6S and $C_2H_6S_2$ are presented in Fig. 1E and F. C_2H_6S, an important oxidation product of $C_2H_6S_2$, belongs to the chemical class of sulfur-containing organic compounds [32]. In the non-aerated control, both compounds persisted until day 20, while aeration significantly enhanced their removal. Notably, at a moderate aeration intensity of 2.3 mL/g(TS)/d on day 10, an 80.6% decrease in $C_2H_6S_2$ concentration was accompanied by a 45.2% increase in C_2H_6S, directly confirming the aerobic conversion of $C_2H_6S_2$ to C_2H_6S in the system. When aeration intensity reached ≥9.3 mL/g(TS)/d, both sulfides were simultaneously and completely removed by day 15. These results indicate that low-intensity aeration primarily regulates odor through structural transformation (malodorous thiol → malodorous disulfide), whereas high-intensity aeration achieves odor elimination via complete mineralization (sulfide → sulfate). The effective odor eradication under high aeration may be attributed to the enrichment of

obligate sulfur-oxidizing bacteria capable of fully oxidizing organic sulfur [33]. The observed patterns of C_2H_6S and $C_2H_6S_2$ thus reveal the profound regulatory effect of aeration on sulfur metabolic pathways.

CS_2 removal exhibited distinct responses to aeration intensity (Fig. 2A). In the unaerated control, CS_2 concentration declined gradually, whereas all aeration treatments markedly accelerated its removal. On day 5, CS_2 levels decreased by 41.3% and 42.6% in the 4.6 and 13.9 mL/g(TS)/d groups, respectively, compared to the control. Moderate aeration (≥4.6 mL/g(TS)/d) reduced CS_2 concentrations to near-zero between days 15-20. Notably, the threshold for complete CS_2 removal (4.6 mL/g(TS)/d) was lower than that required for complete mineralization of C_2H_6S and $C_2H_6S_2$ (≥9.3 mL/g(TS)/d), suggesting that CS_2 may be more susceptible to oxidative degradation [34]. This differential sensitivity highlights the potential for implementing tiered control strategies targeting specific malodorous sulfides.

3.1.3. Integrated odor assessment

Odor pollution is a widespread environmental issue with a subjective nature. Trace sulfur- and nitrogen-containing volatiles can trigger discomfort and complaints [18]. In the European Union (EU), odor emissions from livestock and anaerobic digestion are regulated under the Industrial Emissions Directive (2010/75/EU) via site permits and best available techniques [35]. Odor was quantified by dynamic olfactometry (EN 13725) and expressed in ouE/m^3, where 1 ouE/m^3 is defined as the dilution-to-threshold ratio at the human perception threshold. Residential/rural limits are 5-10 ouE/m^3 [12]. Thus, odor removal evaluation requires whole-sample holistic assessment, not just individual compound quantification.

Changes in overall odor concentration are presented in Fig. 2B. Initial variations may be attributed to the physical stripping effect induced by higher aeration intensities. After system stabilization on day 5, the lowest odor concentration was observed in the 13.9 mL/g(TS)/d treatment. Notably, odor concentration did not exhibit a simple linear decrease with increasing aeration intensity. Instead, the low-intensity groups (2.3-4.6 mL/g(TS)/d) displayed lower odor levels than the moderate-intensity groups (5.8 and 9.3 mL/g(TS)/d) after day 5, suggesting that suboptimal aeration may exacerbate odor formation. Although major odorants (H_2S, NH_3, C_3H_9N) were suppressed in aerated groups, overall odor is influenced by a mixture of compounds such as phenols, aldehydes, and intermediate sulfides [36]. Moderate aeration may therefore facilitate the accumulation of certain non-target malodorous intermediates. Based on integrated odor and key gas results, aeration intensities of 4.6 and 13.9 mL/g(TS)/d demonstrated relatively better overall odor reduction.

3.1.4. Methane production during micro-aeration pretreatment

CH_4 production during micro-aeration pretreatment is shown in Fig. 2C. At the onset of pretreatment (day 0), CH_4 yields in several aerated groups (3.5, 5.8, 9.3, and 13.9 mL/g(TS)/d) were slightly higher than in the non-aerated control. This transient increase may be attributed to improved mass transfer or suppression of competing microbial groups such as sulfate-reducing bacteria [37]. However, as pretreatment progressed, the inhibitory effect of aeration became predominant. A critical threshold was observed at 4.6 mL/g(TS)/d, with CH_4 production decreasing by 71.7% relative to the initial value by day 5, indicating substantial compression of anaerobic niches. At higher intensities (9.3 and 13.9 mL/g(TS)/d), methanogenesis was completely inhibited, and no CH_4 was detected after day 15, reflecting a systemic metabolic shift toward aerobic/facultative oxidative pathways.

These dynamics illustrate the gradient inhibitory effect of oxygen on obligate anaerobic metabolism. The synchronous reduction in CH_4 and malodorous gases reveals an aeration-driven transition from anaerobic fermentation producing reduced end-products such as CH_4 and H_2S, to aerobic/facultative oxidation, yielding oxidized products including CO_2 and sulfate [38].

3.2. Organic matter characteristics and physicochemical properties of pig manure

3.2.1. Dry matter loss

The temporal variation in dry matter mass and the final loss rates under different aeration intensities are shown in Fig. 2D. The initial dry matter mass was 1005.09 ± 4.10 g. After 20 days of pretreatment, all treatment groups exhibited dry matter loss. The unaerated control group showed a natural loss rate of 19.0%. Aeration significantly increased this loss, with rates positively correlated with aeration intensity, loss rates were 24.5%, 23.9%, 28.9%, 34.0%, 36.2%, and 38.8% for the 2.3, 3.5, 4.6, 5.8, 9.3, and 13.9 mL/g(TS)/d groups, respectively. In the highest-intensity group (13.9 mL/g(TS)/d), only 615.33 g of dry matter remained. This dry matter loss mainly results from the aerobic mineralization of labile organic carbon to CO_2 [39]. Moreover, the progressive mass loss observed with increasing aeration intensity indicated that oxygen supplementation promotes carbon mineralization to CO_2. These results clearly indicate that aeration pretreatment accelerates the aerobic mineralization and depletion of organic matter, which may reduce the total substrate available for subsequent anaerobic methanogenesis.

3.2.2. Dynamics of physicochemical parameters

DO variations are presented in Fig. 3A. The DO level in the system was not a simple linear function of aeration intensity but resulted from the dynamic balance between microbial oxygen consumption and physical oxygen supply, indirectly reflecting microbial metabolic activity [37]. In the unaerated control group, DO continuously declined due to microbial consumption and was nearly depleted by day 20. In aerated groups with intensities ≥4.6 mL/g(TS)/d, DO increased rapidly after day 5 and remained elevated, indicating sufficient oxygen supply to support vigorous metabolism by aerobic and facultatively anaerobic microorganisms. Conversely, in low-intensity groups (2.3 and 3.5 mL/g(TS)/d), DO rose slowly and began to decrease after days 15 and 10, respectively, suggesting that the oxygen supply rate was insufficient to meet microbial demand, thereby limiting aerobic processes such as nitrification and sulfur oxidation. Previous studies have reported that obligate anaerobes require anoxic conditions (DO < 0.1 mg/L), facultative anaerobes grow in the micro-aerobic range (DO concentration was 0.1-0.5 mg/L), and aerobes/facultative microbes thrive under aerobic conditions (DO > 0.5 mg/L) [40]. In this study, during zero-aeration pretreatment, DO dropped from 0.2 mg/L to 0, supporting the growth of both facultative and obligate anaerobes. At aeration intensities of 2.3-3.5 mL/g(TS)/d, DO ranged from 0.1 to 0.4 mg/L, providing micro-aerobic conditions conducive to facultative anaerobes. At aeration intensities ≥4.6 mL/g(TS)/d, DO exceeded 0.5 mg/L, creating aerobic conditions that favored aerobes and facultative microbes and potentially inhibited the growth of obligate anaerobes. These findings explain, from a DO availability perspective, why aeration intensities ≥4.6 mL/g(TS)/d achieve faster and more effective suppression of malodorous gas generation.

The ORP changes are shown in Fig. 3B. ORP serves as an electrochemical indicator of the oxidative or reductive capacity within a fermentation system [41]. In the unaerated control group, ORP decreased to −288.53 mV by day 20, reflecting a strongly reducing environment conducive to obligate anaerobes. In contrast, ORP in all aerated groups remained between −152.80 and −85.20 mV, a range more favorable for facultative anaerobes and certain anaerobic microorganisms. The elevated ORP indicates a metabolic shift from obligate anaerobic to facultative/micro-aerobic conditions. This change directly influences microbial community structure and represents a key environmental driver in redirecting odor-related metabolic pathways [42].

pH values in all treatment groups increased gradually from an initial level around 6.0 (Fig. 3C). The low initial pH was attributed to the accumulation of volatile fatty acids (VFAs) from organic matter hydrolysis and acidification. Subsequent pH rise was associated with oxidative consumption of VFAs by aerobic/facultative microorganisms and the

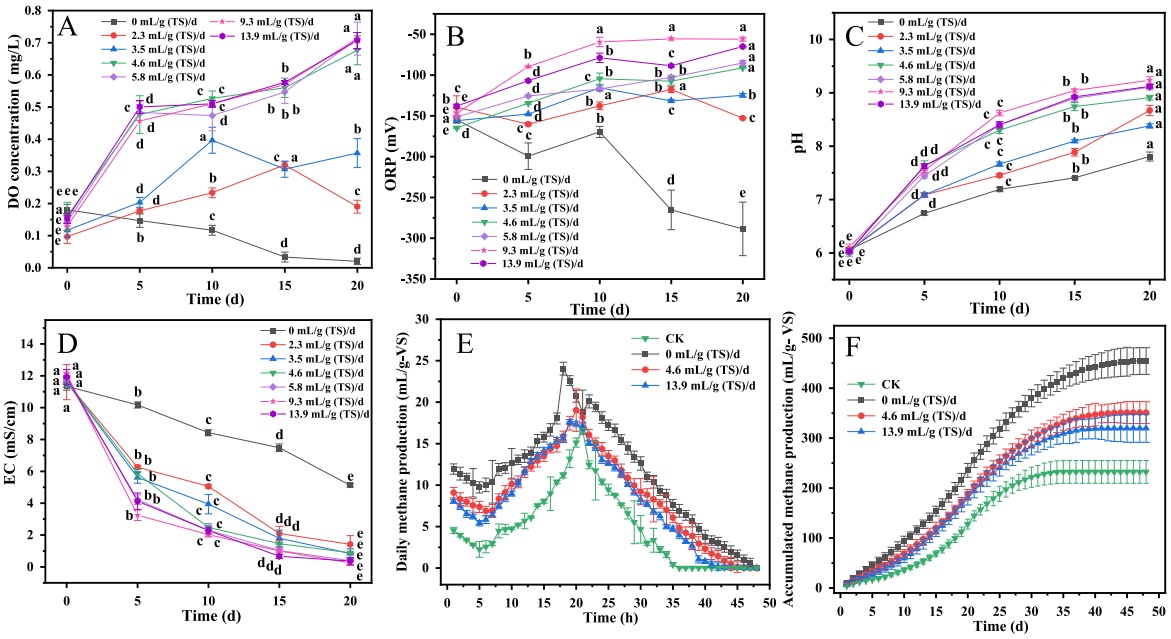


Fig. 3. Effect of micro-aeration on physicochemical properties (A, B, C and D) and methane production (E and F) of pig manure.

neutralization effect of NH_3/NH_4^+ released during protein degradation [6]. Higher aeration intensities corresponded to faster pH increases, consistent with more pronounced organic matter mineralization (as reflected in dry matter loss) and likely accelerated ammonia dynamics.

EC variations are presented in Fig. 3D. EC decreased gradually over time, indicating microbial assimilation and uptake of free ions such as NH_4^+, K^+, and PO_4^{3-} [37]. This process was significantly accelerated by aeration. By day 5, EC in aerated groups was reduced by 38.4-68.1% compared to the control. By day 20, EC in high-intensity aeration groups (≥5.8 mL/g(TS)/d) approached 0 mS/cm. These results demonstrate that aeration strongly stimulates microbial activity and enhances the biological immobilization of nutrients.

3.3. Methane production during anaerobic digestion

The CH_4 production from pig manure during 50-day AD following different pretreatments is shown in Fig. 3E. Daily CH_4 profiles of all treatment groups exhibited typical digestion phases: lag, rising, and declining. Notably, although the unaerated pretreatment group (A0) (454.4 mL/g VS) showed higher cumulative CH_4 production, likely due to its lower ORP promoting early methanogen colonization, untreated raw manure (CK) yielded the lowest cumulative CH_4 output (232.4 mL/g VS). This suggests that pretreatment modifies substrate characteristics through initial microbial metabolism, thereby enhancing subsequent AD performance [43]. However, cumulative CH_4 production from manure pretreated at 4.6 (351.0 mL/g VS) and 13.9 mL/g(TS)/d (319.7 mL/g VS) were lower than those of the unaerated pretreatment group (A0). This contrasts with several reports indicating that in situ micro-aeration enhances CH_4 yield [44]. In this study, AD was conducted in sequential batch mode following pretreatment. High-intensity aeration induced substantial organic matter mineralization during pretreatment (dry matter loss up to 38.8%) and elevated the initial ORP of the substrate upon entry into the anaerobic reactor. Together, these factors reduced the total carbon available for methanogenesis and slowed anaerobic startup [45]. Therefore, although aeration may optimize microbial functionality, irreversible carbon loss emerges as the dominant constraint on overall CH_4 production.

An aeration intensity of 4.6 mL/g(TS)/d exhibited the relatively optimal balance between malodorous gases reduction and energy recovery. The maximum daily gas production rate and cumulative CH_4 yield achieved at this intensity were significantly higher than those from the 13.9 mL/g(TS)/d pretreatment and from untreated raw manure (CK). Although its cumulative CH_4 output was slightly lower than that of the unaerated pretreatment group (A0), it simultaneously delivered substantial malodorous gases mitigation. These results underscore the inherent challenge of simultaneously maximizing malodorous gases control and CH_4 recovery within a single process. Thus, 4.6 mL/g(TS)/d can be regarded as an optimal operational point, effectively controlling malodorous gases while retaining high methanogenic potential within an acceptable range of carbon loss. Excessively high aeration intensities, such as 13.9 mL/g(TS)/d, are economically inefficient unless complete malodorous gases elimination is the sole and absolute objective. Furthermore, micro-aeration pretreatment outperforms the co-digestion of carbon-rich substrates by eliminating the need for external carbon sources, enabling real-time flexible control, enhancing methane production, and reducing emissions of malodorous gases [46]. However, micro-aeration pretreatment does not modify the overall C/N ratio and demands stringent aeration control to prevent the suppression of methane production. Nevertheless, micro-aeration pretreatment is particularly advantageous for intensive pig farms that lack accessible crop residues or other carbon-rich wastes.

3.4. Microbial community structure and diversity

3.4.1. α- and β-diversity

High-throughput sequencing was employed to analyze micro-aeration-induced shifts in microbial community, thereby revealing the mechanisms for odor release and methane production. The Coverage index exceeded 0.999 for all samples, confirming that sequencing depth was sufficient to represent the true community composition. As shown in Table S2, species richness indices (ACE, Chao1, and Sobs) followed the order A13.9 > A4.6 > A0 > CK, indicating that aeration pretreatment, especially at higher intensities, significantly expanded the microbial species pool. However, group A4.6 exhibited the highest Shannon index, reflecting greater community evenness and integrated diversity. Similarly, the Simpson index was significantly higher in groups A0 and A4.6 than in CK and A13.9 (Fig. S3). These results suggest that moderate aeration (4.6 mL/g(TS)/d) promoted a more evenly distributed and structurally stable community, which may correlate with its optimized system performance.

Principal coordinate analysis (PCoA) revealed clear separation in microbial community structure among treatment groups (PERMANOVA, $p < 0.05$; Fig. 4A). The CK group was distinctly separated from all aerated groups (A0, A4.6, A13.9) along the first principal axis, indicating that aeration was the primary driver of community differentiation. Notably, the A0 and A4.6 groups clustered closely in ordination space, whereas the A13.9 group was distinctly separated (Fig. 4B). This spatial pattern suggests that community structure undergoes a qualitative shift when aeration intensity exceeds 4.6 mL/g(TS)/d.

3.4.2. Microbial community composition

Venn diagram analysis revealed that aeration pretreatment significantly altered species composition (Fig. 4C and D). The number of core amplicon sequence variants (ASVs) shared among the four treatment groups was very limited (32 at the genus level and 17 at the species level), whereas each group contained a high number of unique ASVs. In particular, the A13.9 group exhibited substantially more unique ASVs (98 at genus level, 492 at species level), confirming the strong community remodeling effect of aeration pretreatment [33].

At the phylum level, all groups were dominated by Bacillota and Bacteroidota (Fig. 5A). As aeration intensity increased, clear functional differentiation was observed. At class, order, family, and species levels, tolerant anaerobic taxa, such as Bacilli and Syntrophomonadales dominated hydrolysis and acidification processes in the A0 and A4.6 groups. In contrast, high-intensity aeration (A13.9) enriched a distinct syntrophic consortium comprising micro-aerobic/aerobic metabolic bacteria (such as Gammaproteobacteria) alongside efficient methanogens (Methanosarcinales), collectively driving efficient organic matter conversion and CH_4 generation (Fig. S4–S7) [47].

At the genus level, syntrophic groups such as *Syntrophomonas* were significantly enriched in the A0 and A4.6 treatments (Fig. 5B). These taxa play key roles in degrading long-chain fatty acids and supplying substrates to methanogens, which may explain the higher CH_4 production potential observed in these groups [48]. In the A13.9 group, although methanogenic archaea (for example, *Methanothrix*) were also enriched, the relative abundance of taxa associated with aerobic/facultative metabolism increased concurrently. The pretreatment enhanced overall community metabolic activity and increased the generation of methanogenic precursors to promote the enrichment of methanogenic archaea. This changes further supporting the conclusion that the decline in total methane production is largely due to organic matter loss.

On the other hand, the microbial community structure shaped by micro-aeration pretreatment remained largely throughout anaerobic digestion. This retention indicated that key aerobic and facultative aerobic microbes contribute to further elucidate the mechanism of inhibitory odor gas formation. For example, unclassified_f_*Bacillaceae* and unclassified_c_*Gammaproteobacteria* were enriched in group A13.9. unclassified_c_*Gammaproteobacteria* performed desulfurization and denitrification by reducing sulfate/thiosulfate and participating in nitrate reduction, thereby lowering the accumulation of H_2S and nitrogenous odorants [17]. Taxa within unclassified_f_*Bacillaceae* carried out sulfur oxidation and nitrification/denitrification, converting H_2S to sulfate and ammonia to nitrate or N_2 under aerobic conditions [5]. Therefore, micro-aeration pretreatment can improve the methane production efficiency and control the odor gas production by reshaping the microbial community to improve the metabolic capacity of the system.

3.5. Distribution and phylogenetic relationships of key taxa

The differentiation described above was further confirmed by genus-level heatmap clustering (Fig. 6A). The CK group formed a distinct cluster, characterized by taxa such as intestinal *lactobacilli*, while all aerated groups clustered together. Within this aerated cluster, groups A0 and A4.6 formed a sub-cluster due to their high abundance of key functional bacteria including *Syntrophomonas*, reinforcing the functional similarity between these two communities. Group A13.9 branched separately and exhibited a distinct microbial profile, dominated by genera such as *Candidatus_Cloacimonas*, which are closely associated with anaerobic metabolic processes including hydrolytic fermentation, syntrophic oxidation, and acetoclastic methanogenesis [49].

In the A4.6 and A13.9 groups, microbial taxa including

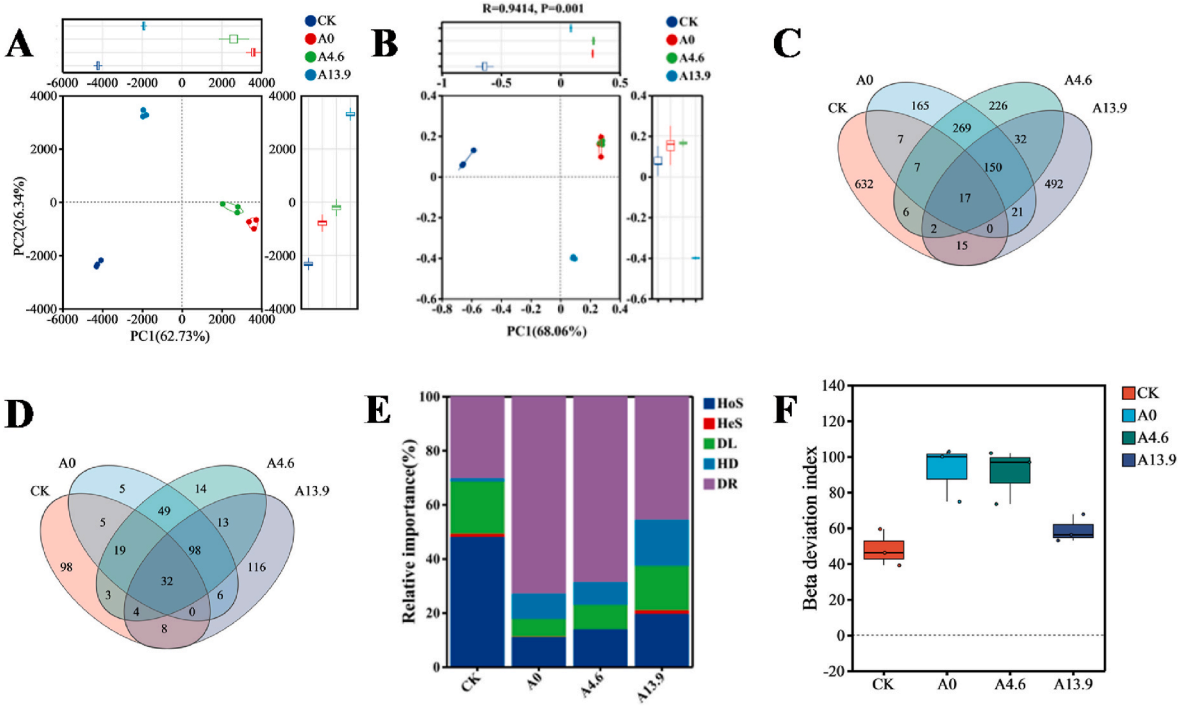


Fig. 4. The PCA (A), PCoA (B), Venn diagram (C and D), iCAMP (E) and Beta-deviation (F) index of microbial community during anaerobic digestion of pig manure after micro-aeration pretreatment.

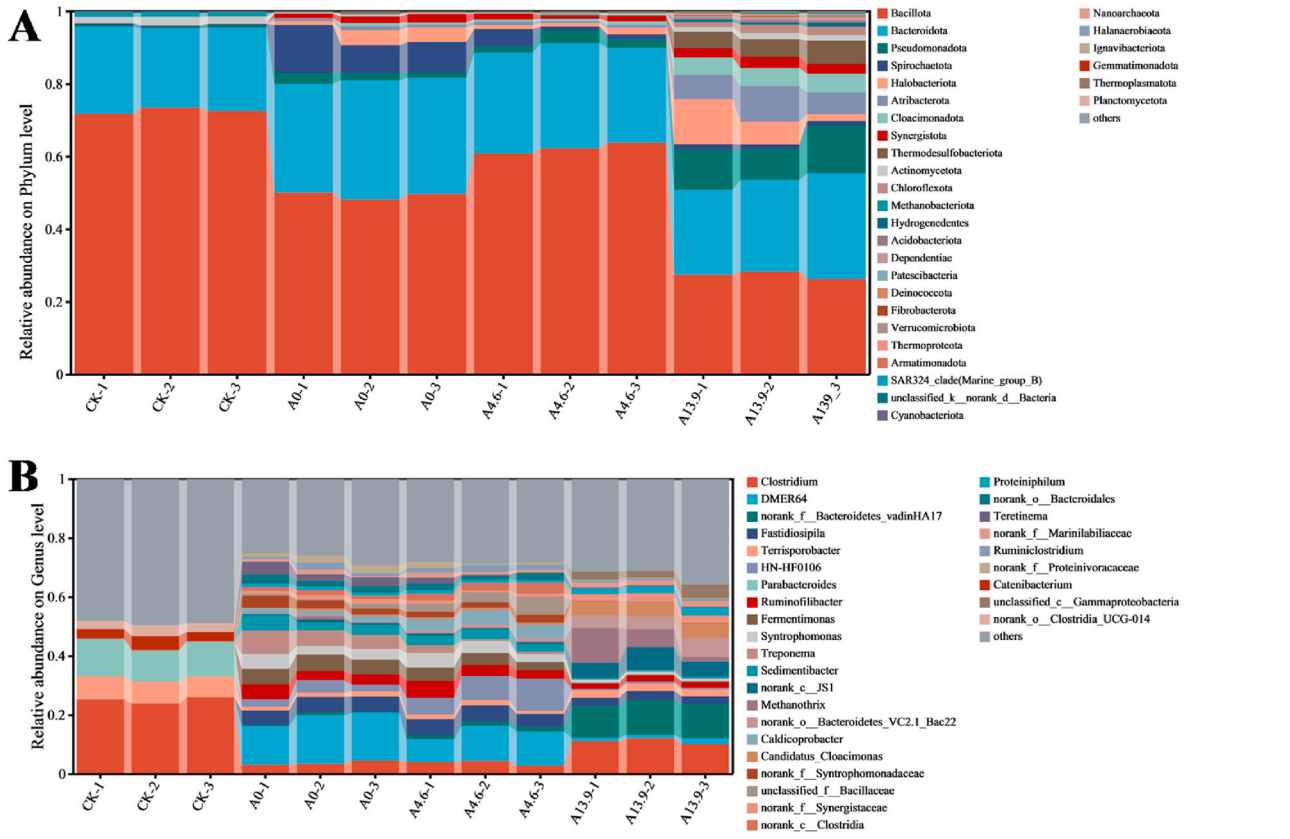


Fig. 5. Effects of micro-aeration pretreatment on the phylum (A) and genus (B) levels of microbial communities during anaerobic digestion of pig manure.

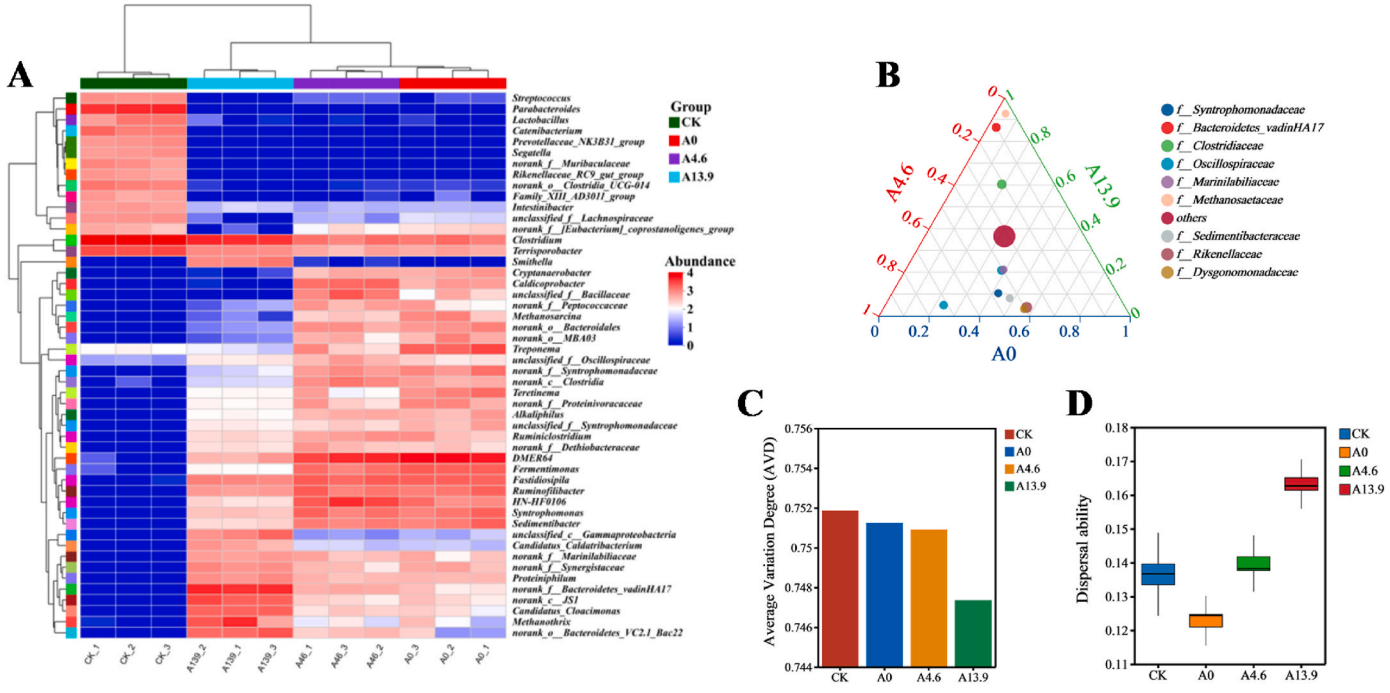


Fig. 6. Microbial gender-level clustering heat map (A), ternary phase diagram (B), AVD (C) and diffusion capacity (D) during anaerobic digestion of pig manure after micro-aeration pretreatment.

Proteiniphilum, *Ruminiclostridium*, *Sedimentibacter*, norank_c_*Clostridia*, norank_f_*Peptococcaceae*, norank_f_*Synergistaceae*, *Syntrophomonas*, and *Methanosarcina* were significantly enriched under micro-aeration pretreatment (Fig. 6A). These microorganisms form a synergistic functional network that accelerated the removal and attenuation of key malodorous compounds. For instance, *Proteiniphilum* and certain *Ruminiclostridium* contributed to desulfurization and denitrification pathways, effectively converting sulfur-containing compounds (H_2S and

organic sulfides) and mitigating ammonia release [15]. Meanwhile, *Sedimentibacter* and *Syntrophomonas* supported the rapid turnover of volatile fatty acids and sulfur intermediates, reducing the accumulation of odor precursors [20]. Collectively, this structured microbial consortium facilitated by micro-aeration pretreatment ensured efficient odor control.

The ternary plot illustrates taxon-specific responses to aeration intensity (Fig. 6B). Key functional bacteria such as *Oscillospiraceae* were predominantly distributed between groups A0 and A4.6, whereas *Bacteroidetes_vadin* HA17 and *Methanothrix* were positioned closer to groups A4.6 and A13.9. This pattern suggested that an aeration intensity of A4.6 may represent an ecological threshold capable of simultaneously enriching hydrolytic-acidogenic bacteria and key methanogenic taxa, thereby facilitating functional coupling.

Phylogenetic analysis revealed that key genera enriched under different aeration regimes were distributed across divergent branches of the phylogenetic tree (Fig. S9D). This indicates that aeration selected microorganisms based on functional traits such as oxygen tolerance and syntrophic capability, rather than specific evolutionary lineages, reflecting the convergent functional adaptation driven by environmental selection.

3.6. Integrated microbial assembly and functional adaptation under micro-aeration

Micro-aeration pretreatment systematically restructured microbial community assembly and functional interaction networks (See supplementary material Text 1-5). Specifically, infer phylogenetic-bin-based null model (iCAMP) analysis revealed that moderate aeration (4.6 mL/g(TS)/d) maximized stochastic drift (61.5%) and community stability (Fig. 4E–F). In contrast, high aeration re-imposed selective pressure, leading to functional simplification (Fig. 6C–D). Moreover, compared with the CK group, generalist species dominated all groups (>80%) after aeration at 4.6 mL/g(TS)/d, while the proportion of neutral species decreased from 6.6% to below 3.5%, enhancing community stability and functional efficiency (Fig. S8). Key taxa succession was traced from fermentative (CK) to syntrophic (A0/A4.6) and ultimately to methanogenic communities (A13.9) using linear discriminant analysis effect size (LEfSe) analysis (Fig. S9A–S9B). Statistical analysis revealed 338 genera exhibiting significant abundance differences ($p < 0.01$, Fig. S9), with a higher degree of compositional similarity between A0 and A4.6 (Fig. S10). After aeration pretreatment, *Methanosarcina* and *Methanothrix* established a cooperative network, with *Methanothrix* accounting for 20% relative abundance in A4.6. This network reduced odor precursor and maintained methanogenic functional redundancy [37].

PICRUSt2-based functional prediction revealed up-regulation of sulfur oxidation, sulfate reduction, nitrogen fixation, and C1 compound degradation in group A4.6 compared with the CK group. These functional enhancements contributed to effective odorant removal [38]. In the A4.6 group, carbon flux was redirected and enhanced via fatty acid β-oxidation, the TCA cycle, and acetyl-CoA synthesis. These metabolic shifts facilitated volatile fatty acid degradation and augmented the availability of methanogenic substrates [43]. This functional synergy enabled at aeration of 4.6 mL/g(TS)/d to achieve optimal odor control and methane production.

4. Conclusions

Micro-aeration reduced the concentrations of the eight target odorants (NH_3, C_3H_9N, H_2S, CH_4S, C_2H_6S, $C_2H_6S_2$, CS_2) by 14.0%-100%. Low-intensity aeration (2.3-3.5 mL/g(TS)/d) only transformed organic sulfides, whereas high-intensity aeration (≥9.3 mL/g(TS)/d) achieved complete removal of sulfur-containing compounds. Moreover, micro-aeration significantly enhanced methane production by 37.6%-95.5% compared with the control. However, elevated aeration intensity concurrently accelerated substrate carbon mineralization, as evidenced by an increase in dry matter loss from 19.0% in the control to 38.8% at the maximum aeration rate (13.9 mL/g(TS)/d). Furthermore, microbial analysis revealed that micro-aeration enriched facultative taxa (e.g., *unclassified_f_Bacillaceae* and *Gammaproteobacteria*) and up-regulated pathways related to sulfur oxidation, nitrogen fixation, fatty acid β-oxidation, and acetyl-CoA synthesis, promoting odor elimination. Moderate aeration (4.6 mL/g(TS)/d) enriched *Syntrophomonas*, *Methanothrix*, and *Methanosarcina*, while enhancing sulfur oxidation and fatty acid conversion potentials that promoted methanogenesis. Overall, 4.6 mL/g(TS)/d was identified as the optimal aeration intensity, achieving the balance between odor removal and methane production. These findings provide theoretical support and operational parameters for regulating anaerobic digestion of pig manure.

CRediT authorship contribution statement

Hong Zhao: Conceptualization, Investigation, Methodology, Writing – original draft, Writing – review & editing. **Xiao-Bo Zhou:** Conceptualization, Investigation, Methodology. **Shu-Han Zhang:** Conceptualization, Investigation, Methodology. **Fan Liu:** Conceptualization, Investigation, Methodology. **Teng-Xia He:** Funding acquisition, Project administration, Resources, Writing – review & editing. **Guang-Liang Tian:** Conceptualization, Investigation, Methodology, Resources. **Xi-Wen Liang:** Conceptualization, Investigation, Methodology. **Man-Man Zhang:** Investigation. **Yi-Xiao Liao:** Investigation.

Declaration of competing interests

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Tengxia He reports financial support was provided by Key Project of Guizhou Provincial Basic Research Program (Natural Science) (QKHJC-ZD [2026] 033), Guizhou Province High-Level Talent Training Support Project in the Field of Scientific and Technological Innovation (Qian Ke Xie KJLYRC-[2026]077) and Scientific Research Innovation Team Project of Guizhou University [No. Guidakechuangtuan (2024) 06]; Postgraduate Research Fund of Guizhou Province (No. 2024YJSKYJJ117); Innovative Training Program for college Students of Guizhou University (No. X2025106570086); Student Research Training Program of Guizhou University (No. 2025SRT466 and 2025SRT483). If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by Key Project of Guizhou Provincial Basic Research Program (Natural Science) (No. QKHJC- ZD [2026] 033), Guizhou Province High-Level Talent Training Support Project in the Field of Scientific and Technological Innovation (No. Qian Ke Xie KJLYRC-[2026]077) and Scientific Research Innovation Team Project of Guizhou University [No. Guidakechuangtuan (2024) 06]; Postgraduate Research Fund of Guizhou Province (No. 2024YJSKYJJ117); Innovative Training Program for college Students of Guizhou University (No. X2025106570086); Student Research Training Program of Guizhou University (No. 2025SRT466 and 2025SRT483).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.renene.2026.126304.

Data availability

Data will be made available on request.

References

[1] I.M. Aroh, G. McCutcheon, B.P. Macartan, R. Kunes, T.P. Curran, L. Clarke, Monitoring ammonia emissions in pig facilities: a comparative review of measurement technologies, monitoring protocols, and technology decision-support framework, Comput. Electron. Agric. 241 (2026), https://doi.org/10.1016/j.compag.2025.111238.

[2] X.N. Li, Y.B. Zhou, Y.L. Su, D. Wu, M. Zhan, B. Xie, Microbial mechanisms enhancing humification and nitrogen retention during co-composting of black soldier fly frass and pig manure, Waste Manage. (Tucson, Ariz.) 208 (2025), https://doi.org/10.1016/j.wasman.2025.115146.

[3] W.H. Liao, M.L. Liu, Y. Bian, W.J. Wang, Z.L. Gao, Towards greenhouse gases mitigation for liquid pig slurry management with solid-liquid separation technologies, J. Environ. Manag. 395 (2025), https://doi.org/10.1016/j.jenvman.2025.127786.

[4] M.J. Chen, Z. Cao, B.Y. Jing, W.J. Chen, X. Wen, M. Han, Y. Wang, X.D. Liao, Y. B. Wu, T. Chen, The production of methyl mercaptan is the main odor source of chicken manure treated with a vertical aerobic fermenter, Environ. Res. 260 (2024), https://doi.org/10.1016/j.envres.2024.119634.

[5] Y.Z. Zhang, X.S. Su, S.Y. Zhao, Y. Chen, Z.C. Lu, G. Wang, Y.J. Li, H.F. Su, Impact of semi-permeable membrane-coupled biochar composting on the mitigation of greenhouse gas emissions and bacterial community, Process Saf. Environ. 202 (2025), https://doi.org/10.1016/j.psep.2025.107838.

[6] V. Pham, N.T. Trung, V. Pham, V. Do, C.Y. Guan, T.K. Lam, P.H. Wu, C.P. Yu, Hydrothermal pretreatment with and without catalysts to improve anaerobic digestion of pig manure, J. Environ. Chem. Eng. 13 (6) (2025), https://doi.org/10.1016/j.jece.2025.119822.

[7] L. Wu, Z.M. Hou, Z.F. Luo, Impacts of microbial competition on underground biomethanation of hydrogen and carbon dioxide: insights from biogeochemical simulations, Renew. Energy 251 (2025), https://doi.org/10.1016/j.renene.2025.123458.

[8] V.A. Lourenco, F.P. Camargo, I.K. Sakamoto, E.L. Silva, M. Varesche, Influence of substrates proportion and concentration on biogas composition and yield in the co-digestion of solid and liquid waste from coffee and swine farms, Renew. Energy 249 (2025), https://doi.org/10.1016/j.renene.2025.123191.

[9] T.T. Hou, Y.Z. Jiao, J.M. Zhao, Z.Y. Zhang, Z.F. Lei, G.Z. Xu, X.H. Pan, P.F. Li, F. Petracchini, C. He, Enhancing methanogenesis efficiency in anaerobic digestion of food waste through incremental salinity and nanobubble utilization to domesticate inoculum: adaptability and optimal inoculation ratios, Renew. Energy 249 (2025), https://doi.org/10.1016/j.renene.2025.123258.

[10] W. Choi, W. Lee, K. Kim, Odor generation pattern of swine manure according to the processing form of feed, J. Anim. Sci. Technol. 66 (1) (2024) 219–231, https://doi.org/10.5187/jast.2023.e48.

[11] H. Zhao, T. He, C. Zheng, C. Wang, H. Liu, Y. Tian, J. Liu, [Research progress of nanoscale zero-valent iron in synergistic combinations of denitrifying bacteria for nitrate removal], Chin. J. Biotechnol. 40 (10) (2024) 3441–3459, https://doi.org/10.13345/j.cjb.240089.

[12] J.R. Zhang, J.G. Li, Z.H. Wu, J. Liu, X.T. You, H. Wang, Z.J. Shen, M.Z. Wang, Effects of mimicking manure removal strategies on ammonia and greenhouse gas emissions in sow pen scale models, Biosyst. Eng. 242 (2024) 169–178, https://doi.org/10.1016/j.biosystemseng.2024.04.016.

[13] R. Kadam, S. Jo, J. Park, Importance of substrate mixture ratio optimization on efficient anaerobic co-digestion of organic wastes generated in livestock sector: insights into process performances and metagenomics, Bioresour. Technol. 441 (2026), https://doi.org/10.1016/j.biortech.2025.133561.

[14] Y.K. Yuan, T.T. Hou, P.F. Li, H. Sun, Y.Z. Jiao, G. Li, C. He, Enhancing semi-continuous anaerobic digestion efficiency with nanobubble water: thermophysical properties optimization and microbial regulation, Renew. Energy 258 (2026), https://doi.org/10.1016/j.renene.2025.124921.

[15] C. Ma, F.R. Dalby, A. Feilberg, L.B. Guldberg, J.J. Yu, U. Marzocchi, S.O. Petersen, Enhanced methane oxidation in surface crusts of manure storages with controlled ventilation, J. Environ. Manag. 393 (2025), https://doi.org/10.1016/j.jenvman.2025.127214.

[16] X.J. Wang, J.T. Li, C.Y. Chen, Z.F. Luo, Y.W. Pang, H.X. Tu, X.J. Lin, C.F. Long, Q. Y. Cai, Z.B. Wei, J.R. Qiu, Optimization of degradation conditions and elucidation of novel biodegradation pathways for sulfamonomethoxine by a novel bacillus strain, Appl. Environ. Microbiol. 91 (9) (2025), https://doi.org/10.1128/aem.01329-25.

[17] Y.Y. Feng, L.S. Wang, Y.H. Liu, H.F. Xie, J. Li, Y.H. Ji, S.Y. He, X. Zhang, H.B. Fu, Y. F. Feng, G. Poinern, L.H. Xue, Synergistic NH3 and N2O mitigation in rice paddies via subcritical thermochemical conversion of livestock manure: unraveling subcritical fecal fertilizer-mediated biochemical pathways, Chem. Eng. J. 524 (2025), https://doi.org/10.1016/j.cej.2025.169184.

[18] Y.X. Li, L.C. Campos, Y.K. Hu, The role of sludge microstructure in enhancing anaerobic digestion: a comprehensive review combining bibliometric and experimental insights, Environ. Dev. Sustain. (2025), https://doi.org/10.1007/s10668-025-06702-6.

[19] H. Zhao, F. Liu, T. He, W. Tian, C. Zheng, Y. Liao, M. Zhang, Y. Tian, H. Liu, Boosting microbial nitrogen removal through green-engineered iron nanocomposites: a sustainable strategy for high-strength inorganic nitrogen wastewater treatment, Environ. Res. 280 (2025) 121967, https://doi.org/10.1016/j.envres.2025.121967.

[20] R. Zhu, D.H. Wang, Y. Zheng, H. Zou, S.F. Fu, Understanding the mechanisms behind micro-aeration to enhance anaerobic digestion of corn straw, Fuel 318 (2022), https://doi.org/10.1016/j.fuel.2022.123604.

[21] J.Y. Zheng, X.P. Wang, Y.B. Zhang, Synergistic promotion of straw anaerobic digestion by a composite microbial consortium and micro-aeration: enhancing lignocellulose decomposition and methanogenesis, Chem. Eng. J. 528 (2026), https://doi.org/10.1016/j.cej.2025.172265.

[22] Y.Z. Qian, S.L. He, C. Rong, Y. Guo, F.Q. Chen, J.H. Shen, Y. Qin, Y.Y. Li, Impact of Ca supplement to the single-stage partial nitritation/anammox-hydroxyapatite process in treating swine manure digestate: deterioration mechanism and recovery strategy, Chem. Eng. J. 522 (2025), https://doi.org/10.1016/j.cej.2025.167428.

[23] H. Zhao, W. Tian, T. He, F. Liu, C. Zheng, M. Zhang, Y. Liao, J. Liu, Enhancing microbial nitrogen removal capacity by modified nanoscale zero-valent iron bio-interface: from performance to mechanism, Chem. Eng. Sci. 322 (2026) 122953, https://doi.org/10.1016/j.ces.2025.122953.

[24] H.A. Mohamed, Z. Rengel, N. Bolan, B.A. Khan, K. Siddique, Z.M. Solaiman, Adsorption of ammonium from anaerobic food waste digestate by pristine and modified eucalyptus biochar for nitrogen fertiliser use, J. Soil Sci. Plant Nutr. 25 (2) (2025) 4531–4551, https://doi.org/10.1007/s42729-025-02414-y.

[25] M. Sobhi, T. Elsamahy, E. Zakaria, M.S. Gaballah, F. Zhu, X. Hu, C. Zhou, J. Guo, S. Huo, R. Dong, Characteristics, limitations and global regulations in the use of biogas digestate as fertilizer: a comprehensive overview, Sci. Total Environ. 957 (2024), https://doi.org/10.1016/j.scitotenv.2024.177855, 177855-177855.

[26] L. Veen, H. van den Brand, A. Oever, B. Kemp, M. Meisenburg, Manure odor profiling for flock-level monitoring on commercial layer pullet farms: vaccination events as a model stressor, Poult. Sci. 104 (2) (2025), https://doi.org/10.1016/j.psj.2024.104681.

[27] H. Zhao, S. Zhang, X. Zhou, F. Liu, T. He, C. Zheng, Y. Liao, L. Lv, X. Zou, X. Wang, L. Yang, M. Zhang, Klebsiella michiganensis BY30: simultaneous phenol degradation and inorganic nitrogen removal by heterotrophic nitrification-aerobic denitrification, Chem. Eng. Sci. 336 (2026) 124531, https://doi.org/10.1016/j.ces.2026.124531.

[28] X. Xiao, K. Kuang, Z.J. Tang, X. Yang, H.W. Wu, Y.Q. Wang, P. Fang, Emission and spatial variation characteristics of odorous pollutants in the aerobic tank of an underground wastewater treatment plant (UWWTP) in southern China, Environ. Pollut. 346 (2024), https://doi.org/10.1016/j.envpol.2024.123631.

[29] P. García, A. Feilberg, L.B. Guldberg, M.J. Hansen, Study on ammonia, methane, and odour emissions dynamics from pig slurry systems in an experimental chamber, Biosyst. Eng. 257 (2025), https://doi.org/10.1016/j.biosystemseng.2025.104237.

[30] Y.Z. Ren, R.F. Zhang, L. Zhang, H.G. Wang, X.Y. Zhang, Z.H. Bai, L. Ma, X. Wang, The characteristics of key odorants from livestock farms and their mitigation potential: a meta-analysis, Atmosphere 16 (9) (2025), https://doi.org/10.3390/atmos16091097.

[31] Y.X. Shi, X.H. Gu, J.C. Song, H. Li, C. Sun, J.H. Wang, J.J. Li, W. Liu, G.X. Bu, H. B. Wang, S.Y. Li, Odor emissions and humification during composting of pig manure and cornstalks: role of mixing proportions, J. Environ. Chem. Eng. 13 (5) (2025), https://doi.org/10.1016/j.jece.2025.117586.

[32] L. He, Z.C. Yang, Y.Y. Xu, X.K. Lu, S. Kaba, C.M. Li, X.L. Yao, D. Xu, Z.L. Yao, Copper doped UiO-66@TiO2 heterojunction for highly efficient photodegradation of odors in broiler houses, J. Clean. Prod. 451 (2024), https://doi.org/10.1016/j.jclepro.2024.142103.

[33] D.N. Ruan, Z. Zhou, H.J. Pang, J. Yao, G. Chen, Z. Qiu, Enhancing methane production of anaerobic sludge digestion by microaeration: enzyme activity stimulation, semi-continuous reactor validation and microbial community analysis, Bioresour. Technol. 289 (2019), https://doi.org/10.1016/j.biortech.2019.121643.

[34] X.Z. Gao, F.Y. Yang, J.W. Cheng, Z.C. Xu, B. Zang, G.X. Li, X.M. Xie, W.H. Luo, Emission of volatile sulphur compounds during swine manure composting: source identification, odour mitigation and assessment, Waste Manage. (Tucson, Ariz.) 153 (2022) 129–137, https://doi.org/10.1016/j.wasman.2022.08.029.

[35] F. Mauro, R. Borghesi, Using citizen science to manage odour emissions in national IED plants: a systematic review of the scientific literature, Atmosphere 15 (3) (2024), https://doi.org/10.3390/atmos15030302.

[36] M.P. Martínez-Hernando, A.G. Alvaro, C.R. Palomar, M. Rodríguez, D. Bolonio, I. D. Crespo, M.J. García-Martínez, M.F. Ortega, Sustainability assessment of a novel vehicular biomethane production from swine manure compared to conventional techniques, Waste Manage. (Tucson, Ariz.) 209 (2026), https://doi.org/10.1016/j.wasman.2025.115218.

[37] J.M. Falk, P. Kjeldsen, C. Buck, C. Scheutz, Treatment of manure methane emissions in a full-scale biofilter by microbial oxidation, Waste Manage. (Tucson, Ariz.) 209 (2026), https://doi.org/10.1016/j.wasman.2025.115214.

[38] X.B. Chen, P.Y. Sun, Z.X. Zhuang, I. Ahmed, L.Z. Zhang, B. Zhang, Control of odorants in swine manure and food waste co-composting via zero-valent iron/H2O2 system, Waste Manage. (Tucson, Ariz.) 174 (2024) 390–399, https://doi.org/10.1016/j.wasman.2023.12.018.

[39] T. Kobayashi, H. Kuramochi, Anaerobic digestate readily reacts with CO2 in pyrolysis to produce inorganic solid residues that exhibit higher microbial enrichment than biochar in anaerobic digestion, Renew. Energy 248 (2025), https://doi.org/10.1016/j.renene.2025.123123.

[40] N. Rahman, C. Eskicioglu, Hydrogen peroxide pretreatment of aqueous phase product of hydrothermal sludge liquefaction for enhanced anaerobic and aerobic biodegradability, Renew. Energy 256 (2026), https://doi.org/10.1016/j.renene.2025.124263.

[41] G. Soggia, A. Goglio, P. Cristiani, I. Luciani, E. Clagnan, F. Adani, Bioelectrochemical protein production valorising NH3-rich pig manure-derived wastewater and CO2 from anaerobic digestion, Renew. Energy 229 (2024), https://doi.org/10.1016/j.renene.2024.120761.

[42] Y.Z. Qian, S.L. He, F.Q. Chen, J.H. Shen, C. Rong, Y. Guo, Y. Qin, Y.Y. Li, Performance stability and recovery strategy of pre-denitrification coupled partial

nitritation/anammox-hydroxyapatite processes for enhanced organic matter, nitrogen and phosphorus removal from swine manure digestate, Bioresour. Technol. 439 (2026), https://doi.org/10.1016/j.biortech.2025.133360.
[43] K.K. Ding, B.C. Wu, Y. Wang, L.J. Xu, M. Liu, J.J. Xiang, Y.D. Chen, L. Gu, J.Z. Li, L. Li, Q. He, S.W. Liu, Study on synergistic effect of carrier combined with micro-aeration on anaerobic digestion of food waste, Chem. Eng. J. 498 (2024), https://doi.org/10.1016/j.cej.2024.155731.
[44] J. Wang, X.D. Wang, C.L. Bian, J.G. Liu, B.Y. Xiao, Effect of aeration pretreatment on anaerobic digestion of swine manure, Environ. Technol. 46 (12) (2025) 2035–2051, https://doi.org/10.1080/09593330.2024.2416094.
[45] I. Zeb, S. Yousaf, M. Ali, A. Yasmeen, A.Z. Khan, J.A. Tariq, Q.B. Zhao, A.M. Abbasi, R. Ahmad, T.M. Khalil, A. Yaqoob, M. Bilal, In-situ microaeration of anaerobic digester treating buffalo manure for enhanced biogas yield, Renew. Energy 181 (2022) 843–850, https://doi.org/10.1016/j.renene.2021.09.089.
[46] S. Jose, B.B. Sajeena, Performance and kinetic study of poultry litter co-digestion with delignified coir pith for biomethane generation, Biomass Bioenergy 199 (2025), https://doi.org/10.1016/j.biombioe.2025.107909.
[47] M. Wang, K. Liu, Q.C. Li, Y.T. Li, H. Peng, Q.J. Wang, M.M. Wang, W.Z. Li, Microbial mechanism for improving biogas production performance from boiling acidified pig manure feedstock, Bioresour. Technol. 419 (2025), https://doi.org/10.1016/j.biortech.2025.132104.
[48] P.Y. Nguyen, A. Ferreira, P.R. Santiago, F. van der Zee, S. Lacroix, J.H. Tian, P. Lens, Biogas upgrading by in-situ biomethanation in high-rate anaerobic bioreactor treating biofuel condensate wastewater, Renew. Energy 248 (2025), https://doi.org/10.1016/j.renene.2025.123113.
[49] Y. Shen, X. Zhang, M. Ye, X.H. Zha, R. He, Effects of Fe-modified digestate hydrochar at different hydrothermal temperatures on anaerobic digestion of swine manure, bioresource, Tech 395 (2024), https://doi.org/10.1016/j.biortech.2024.130393.